

How we're building Marn

A companion app for physical rehab and pain relief, delivered through one-to-one assisted stretching and recovery in studios — tracked daily in an app, so relief is provable, not just felt.

THE ONE-SENTENCE VERSION

Marn is a companion app for physical rehab and pain relief: assisted stretching and physical recovery delivered one-to-one in studios, tracked in an app a member opens every day, so the relief they feel is backed by a number they can watch move over months.

WHY THIS WORKS

Pain relief and physical rehab are what a member is buying; the app is where they live with it, and the studio is where the physical work happens. A member opens the app between visits — to see what hurt before and after, whether it's actually easing, and what to do at home. Every member has a longitudinal record behind that experience: ten joint angles in degrees, captured every six to eight sessions, tied to what was worked and how they slept. That record is what turns "I feel a bit better" into something a member can see and trust, and it's what makes the pain-relief claim provable rather than anecdotal.

THREE SURFACES, ONE BACKEND

SURFACE	WHO	WHERE
Member app	Paying members	iOS, Android, mobile web
Coach console	Flexologists, studio managers	Tablet at the bed, laptop at the desk
Corporate portal	Employer accounts	Desktop web (Phase 4)

POSITIONING — THE WORD WE DON'T USE

Marn is a **wellness and recovery studio**, not a clinic — deliberately, to stay outside Dubai Health Authority clinic licensing. Pain relief is the headline benefit we market; what's restricted is how we describe delivering it. Coaches, not therapists. Assessments and observations, not diagnoses or treatment plans. Measured change ("hip flexion up 9°, reported pain down from 6 to 3"), never a promise to cure. A member reporting a red flag is gated to a physician referral, not waved through — the clearest evidence we aren't practising medicine.

How we're building it

SHAPE — ONE DEPLOYABLE

A single Next.js application serves the member web surface, the coach console, the corporate portal, and the API, backed by one Postgres database. The mobile app is a React Native client against the same API — a separate client, not a separate service. No service mesh, no microservices, no message broker: one engineer cannot operate a distributed system and shouldn't try.



TECHNOLOGY STACK

LAYER	CHOICE
Language	TypeScript, end to end
Web + API	Next.js (App Router), route handlers
Mobile	React Native via Expo
Database	PostgreSQL, Drizzle ORM, migrations checked into the repo
Jobs	pg-boss — a queue inside Postgres, no Redis to operate
Auth	better-auth, self-hosted, phone OTP primary
UI	Material UI, styled only through a theme + tokens — no hardcoded colors or components
Hosting	Vercel + Neon for now; containers in the UAE region before the first corporate contract

DATA TIERING

Health tier — members, assessments, measurements, sessions, flags, check-ins, programmes, scores. UAE-resident, never leaves. **Operational tier** — product analytics, marketing attribution, error telemetry. May live anywhere, but is scrubbed of health data first: error reporting strips request bodies by default, so a member's pain score can never end up sitting in a foreign log aggregator by accident.

WHERE WE ARE TODAY

The repo now has two trees. **prototype/** holds the original no-real-auth build — a working, self-contained reference, frozen on purpose. The **repo root** is the real product: a bare Next.js + MUI shell with no feature code yet, the starting line for everything on the next page, built against real authorization this time, not the prototype's shortcuts.

What it will contain

Every feature below carries a phase marker — **P1** pilot, **P2** launch, **P3** depth, **P4** scale — so build order is never a guess.

MEMBER APP

P1 Readiness screening

Referral gate before anyone books — red flags route to a physician.

P1 Scores & progress

Flexibility, mobility, recovery, consistency — what's working, what's limiting, what hasn't moved.

P1 Body map

Whole-body range-of-motion view, ten measured groups, front and back.

P1 Session history

Plain-language summaries after every visit — the thing the member actually paid for.

P2 Self-service booking

Live equipment/resource availability, not just a coach's calendar.

P2 Home programme

Prescribed stretches between visits, with adherence tracked.

P2 Pre-session check-in

Sleep, pain areas, notes — two taps on a body diagram before a session.

P3 Wearable connection

HRV, sleep, and strain feeding a real recovery score.

P3 Community & milestones

Opt-in friends and shared progress; consent-gated by design.

COACH CONSOLE

P1 Floor schedule

Today's list — times, names, services, at a glance.

P1 Assessment capture

Ten joint angles, manual entry first, BodyMap device second.

P1 Session logging & flags

Modalities, RPE, pain before/after, mandatory member summary, safety flags.

P2 Programme prescription

Assigning the home programme a member sees on their side.

P3 Outcome metrics

A coach's own results as a coaching tool, not a scoreboard.

P3 Capacity & waitlist

Utilisation view and short-notice fill.

CORPORATE PORTAL **P4** & ADMINISTRATION **P2/P3**

Employers see **population trends only** — aggregate, anonymised, with a minimum cohort size so no report can identify an individual. They never see one member's data, and the product says so loudly. Administration covers service/price management, coach roster and permissions, and credit administration with a full audit trail.

How we'll build it over time

Sequenced as vertical slices — each phase ends with something usable on the studio floor, not just foundations.

WE ARE HERE



Phase 4

Phase 1 — Pilot weeks 0–8

GOAL: THE STUDIO CAN OPEN AND OPERATE

Schema and migrations · coach authentication · member roster · readiness screening · assessment capture (manual first) · session logging with mandatory summary · safety flags · scoring engine · read-only member portal · manual booking by staff.

Exit: a coach runs a full day, and every member can see their results, with no spreadsheet in the building.

Phase 2 — Launch weeks 8–14

GOAL: MEMBERS SELF-SERVE

Member authentication · self-service booking on the resource model · payments and the credit ledger · push/WhatsApp notifications · home programmes · pre-session check-in · mobile app (iOS first, Android within three weeks).

Exit: a member joins, books, pays, attends, and reads their results with no staff involved.

Phase 3 — Depth months 4–7

GOAL: RETENTION MECHANICS AND A SECOND SITE

Wearable integrations · consent-gated community · milestones · waitlist and capacity fill · coach-outcome dashboards · multi-site · re-test reminders · referral tracking.

Phase 4 — Scale months 7–12

GOAL: ENTERPRISE AND NEW FORMATS

Corporate portal · offline capture for pod sites · migration to UAE-hosted production · auto-generated summaries · outcome analytics · Arabic and RTL.

SEQUENCING RULES

- Coach console before member app, always — the console is operationally load-bearing, the app is not.
- No feature ships to members before it works on the floor.
- The UAE hosting migration happens before the first corporate contract, not after it's signed.
- Arabic is deferred but never architecturally blocked — strings and layouts are RTL-ready from day one.

Non-negotiables

Five rules that hold regardless of which phase we're in — violating any of them is treated as a blocking defect, not a style note.

Measurement provenance is mandatory

Every measured value carries its source (BodyMap device, coach entry, or member-reported), instrument, protocol version, and who/when. A derived or estimated value never lands in the same column as a measured one — charts must always be able to say where a point came from.

Wellness studio, not a clinic

No diagnosis, no treatment claims, no clinical vocabulary anywhere copy is generated. Pain relief is sold as the outcome, described as measured change — never as a cure. This is what keeps us outside DHA clinic licensing.

Safety flags route to a human

A red-flag screening answer or a coach-raised concern gates the affected activity until a named person clears it. No automatic expiry, no clearing on the member's own say-so.

Authorization is server-side, always

Every read and write is checked against the session's actual role. A member ID supplied by the client is an input to validate, never a fact. A corporate admin sees aggregates — never one member's measurements.

Health data stays in the UAE, and out of logs

The health tier — measurements, flags, identifiers — is UAE-resident and never appears in logs, traces, analytics, or third-party payloads. A new vendor touching member data is a documented decision, not a config change.

NEXT STEPS

Phase 1 build starts at the repo root against this plan. The prototype stays running as a reference for UI shape and journey pacing, but nothing in it — its auth shortcuts, its flat schema — is assumed to carry over as-is. Feature requests land against the phase they belong to; anything pulled forward early gets flagged, not silently built.